

# Tax Incidence

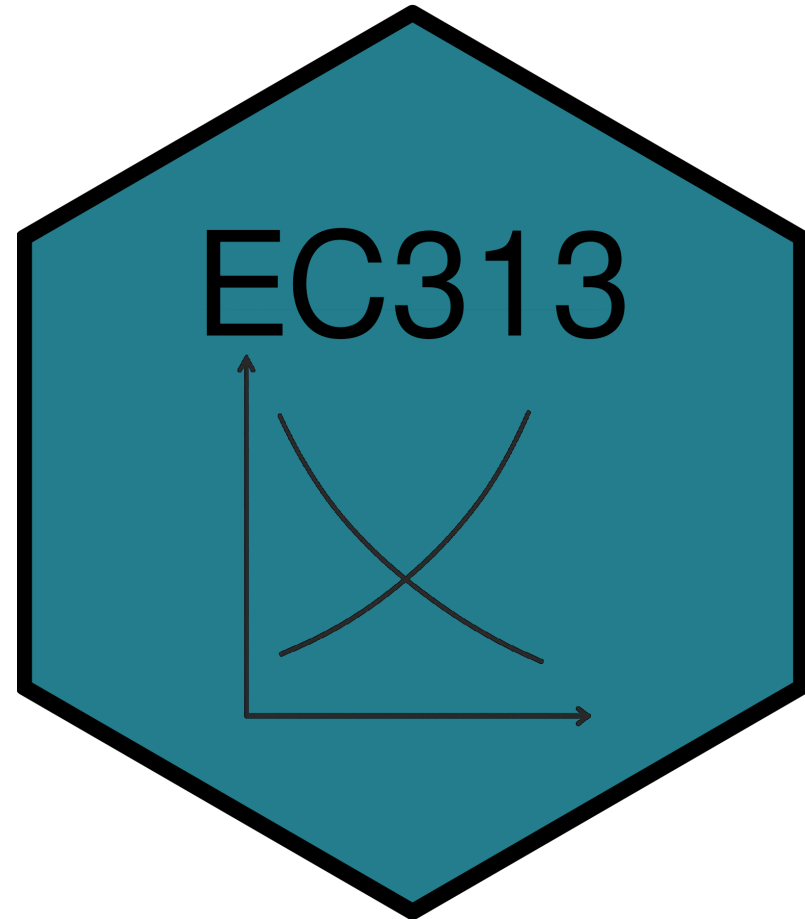
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EC313 - Public Economics: Taxation

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# Goals of This Section

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In this section we will:

- Outline different types of taxes
- Discuss the difference between statutory and economic incidence of a tax
- Show how tax incidence depends on elasticities of supply and demand
- Expand on tax incidence in various markets

# Types of Taxes

# Introduction

- There are many different types of taxes
- Taxes can have different goals
  - Raise revenue for government spending
  - Change behaviour of individuals or firms
  - Redistribute resources
- They can also have different bases, structures, and rates
  - **Base**: what is being taxed (e.g. income, consumption, property)
  - **Structure**: how the tax is applied (e.g. progressive, regressive, flat)
  - **Rate**: how much is being taxed (e.g. percentage, fixed amount)
- Below we cover some of the most common types of taxes

# Taxes on Income

- **Income Tax**: tax on income earned during the year
- Can be levied on individuals or corporations
- For individuals, includes but not limited to
  - Labour market earnings
  - Capital gains
  - Investment income (e.g. from dividends, interest, property)
  - Pensions and retirement income (e.g. RRSPs)
  - Some government benefits (e.g. employment insurance)
  - Other taxable benefits (e.g. premiums paid by employer for group life insurance)

# Taxes on Income

- For corporations, includes but not limited to
  - Active business income from sales of goods and services
  - Investment income
  - Capital gains
  - Some government grants
  - Depends on the size of your business
    - Small businesses pay a lower rate

# Payroll Taxes

- **Payroll Taxes:** taxes levied on employment income
  - Paid by both employers and employees
- Used to finance social insurance and public pension programs
  - Examples: Employment Insurance (EI), Canada Pension Plan (CPP), workers' compensation
- Payroll taxes vary by province
  - Some charge a health tax (e.g. BC Employer Health Tax)
  - Manitoba charges a levy for health and postsecondary education
- These are different from the general income tax

# Consumption Taxes

- **Consumption Tax**: a tax paid on consumption of goods and services
- Taxes generally charged by a seller at point of sale
  - They remit these funds to the government
- Examples:
  - Sales tax (e.g. GST, HST, PST)
  - Excise taxes (e.g. gasoline, alcohol, tobacco)
  - Tariffs (tax on imports)

# Wealth Taxes

- **Wealth Taxes**: taxes on the value of an asset
- Typical wealth taxes include
  - Property tax (tax on value of land/buildings)
  - Estate tax (tax on value of estate at death)
  - General wealth tax (tax on total value of assets owned)
- Estate taxes and general wealth taxes are not used in Canada
- Property taxes are a major source of revenue for municipal governments

# Statutory vs Economic Incidence of a Tax

# Introduction

- The question of who “pays” a tax is more complicated than it seems
- Example: in Canada, there is a federal \$0.10/litre tax on gasoline
  - Gasoline stations include this in their price
  - They remit the tax to the government
  - Does the seller or the consumer pay?
- This section will clarify who pays a tax
- Separate between **statutory** and **economic** incidence of a tax
- My view: this is the most important concept we teach in this program

# Statutory vs Economic Incidence of a Tax

- **Statutory Incidence:** who is legally responsible for paying the tax to the government
  - In the gasoline tax example, the statutory incidence is on the gasoline station
    - They send a cheque to the government
- **Economic Incidence:** the change in real income brought about by the tax
  - In the gasoline tax example, the economic incidence can be shared between the gasoline station and the consumer
    - The gasoline station may receive less revenue per litre sold
    - The consumer may pay a higher price per litre purchased

# Statutory vs Economic Incidence of a Tax

- Example: \$0.10/litre tax on gasoline
  - Before tax is imposed, suppose price is \$1.00/litre
    - Consumers pay \$1.00/litre, gas station receives \$1.00/litre
  - Then government levies \$0.10/litre tax on gasoline station
    - Suppose gasoline station raises price to \$1.10/litre
    - Consumer pays \$1.10/litre
    - Gas station keeps \$1.00/litre, remits \$0.10/litre to government
  - In this case, the **consumer bears the entire economic incidence** of the tax
    - Gasoline station receives same revenue per litre as before tax
    - Consumer pays \$0.10/litre more than before tax

# Statutory vs Economic Incidence of a Tax

- Example 2: \$0.10/litre tax on gasoline
  - Same \$0.10/litre tax on gas station
    - Suppose gas station raises price to \$1.05/litre
    - Consumer pays \$1.05/litre
    - Gas station keeps \$0.95/litre, remits \$0.10/litre to government
  - In this case, the **consumer and gas station share the economic incidence** of the tax
    - Gas station receives \$0.05/litre less than before tax
    - Consumer pays \$0.05/litre more than before tax

# Statutory vs Economic Incidence of a Tax

## Key Insight

Statutory incidence does not determine economic incidence.

- In example, statutory incidence is always on the gas station
- Economic incidence depends on how much of the tax is passed on to consumers
  - Gas station could pass on all, some, or none of the tax to consumers
- To determine economic incidence, we need to look at underlying economic forces

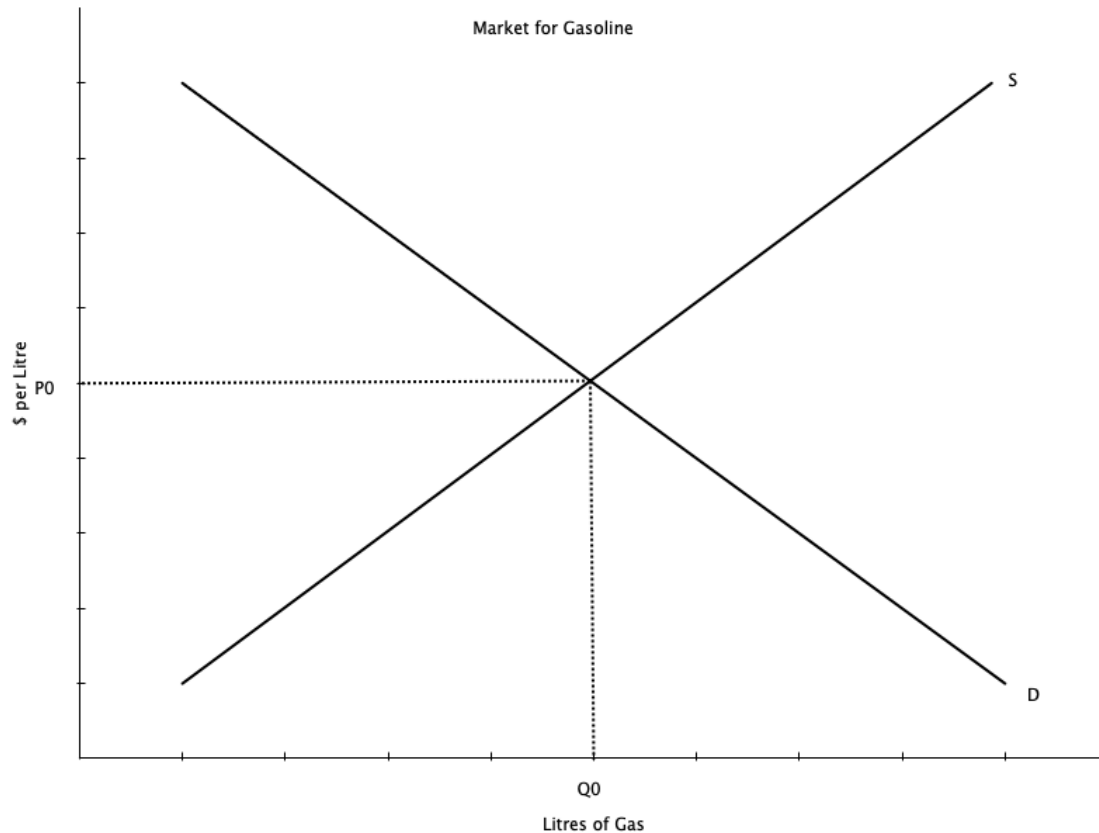
# iClicker Question

A city changes its hotel tax so that guests now pay it at checkout instead of hotels remitting it. The mayor claims this shifts the burden of the tax from hotels to visitors. What should we expect?

- A) Nothing changes — statutory incidence does not determine economic incidence
- B) The burden shifts fully to guests, as the mayor claims
- C) Guests bear the whole burden under either rule
- D) Hotels bear the whole burden under either rule
- E) The burden now depends on who has more bargaining power

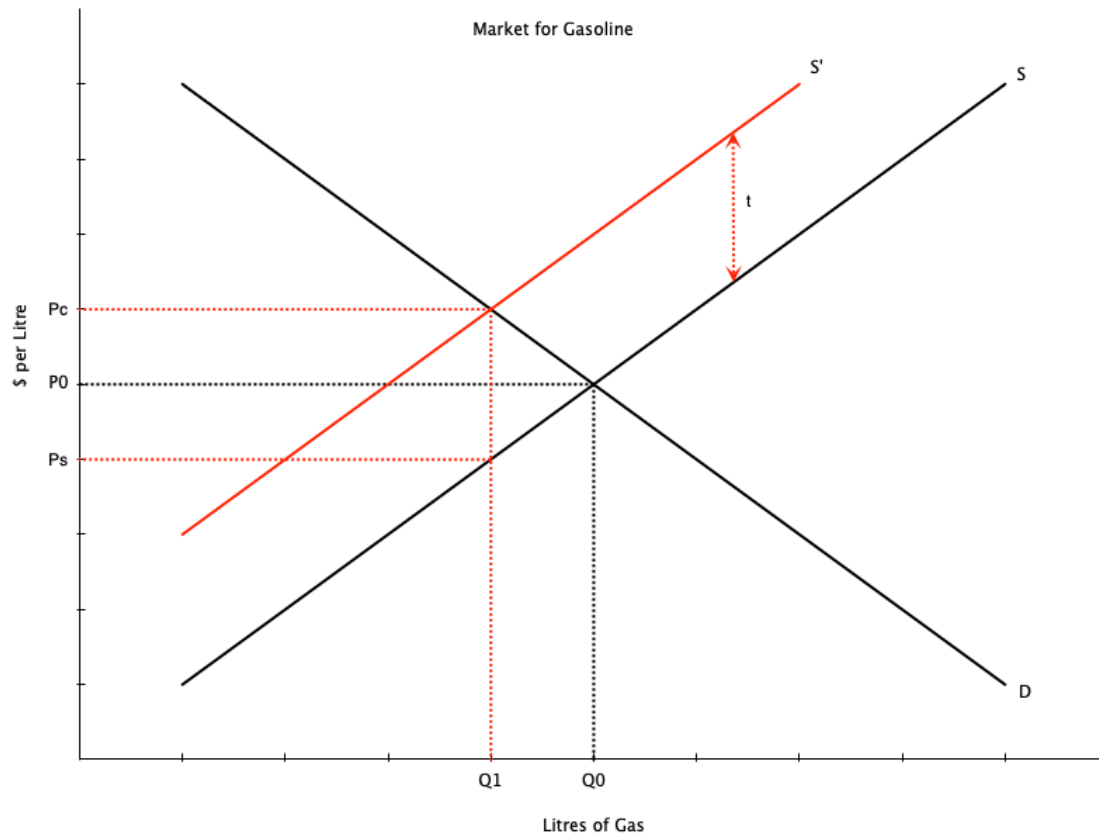
# Commodity Taxes in Partial Equilibrium Models

# Unit Tax on Sellers - Graphical



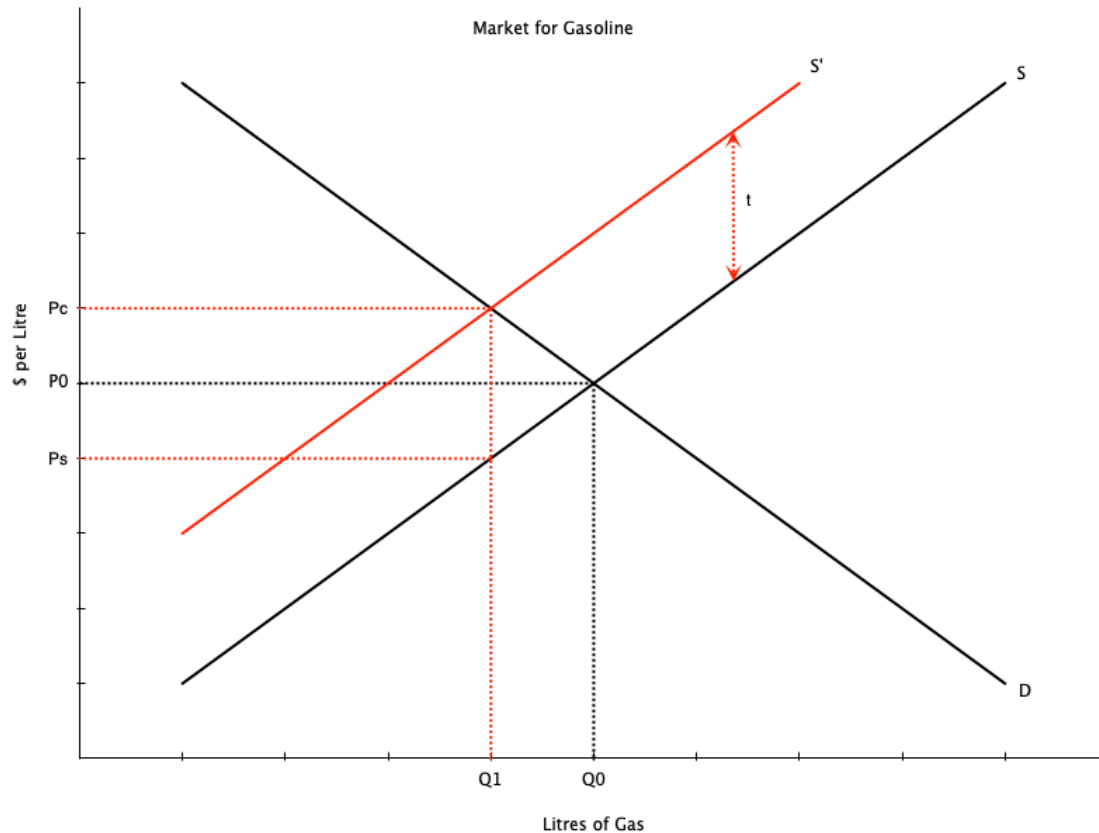
- Take gasoline tax example one more time
- On right is demand and supply of litres of gasoline
- Without tax, price and quantity are determined where demand and supply are equal
  - Price is  $P_0$
  - Quantity is  $Q_0$

# Unit Tax on Sellers - Graphical



- Now government levies a per unit tax  $t$  (e.g. \$0.10/litre) on gasoline
- Tax is levied on **sellers** (statutory incidence)
- This shifts the supply curve up by the amount of the tax
  - New supply curve is  $S'$
  - At each quantity, sellers want to charge  $t$  more to cover the tax
- New equilibrium is where  $S'$  intersects  $D$ 
  - Determines price paid by consumers

# Unit Tax on Sellers - Graphical

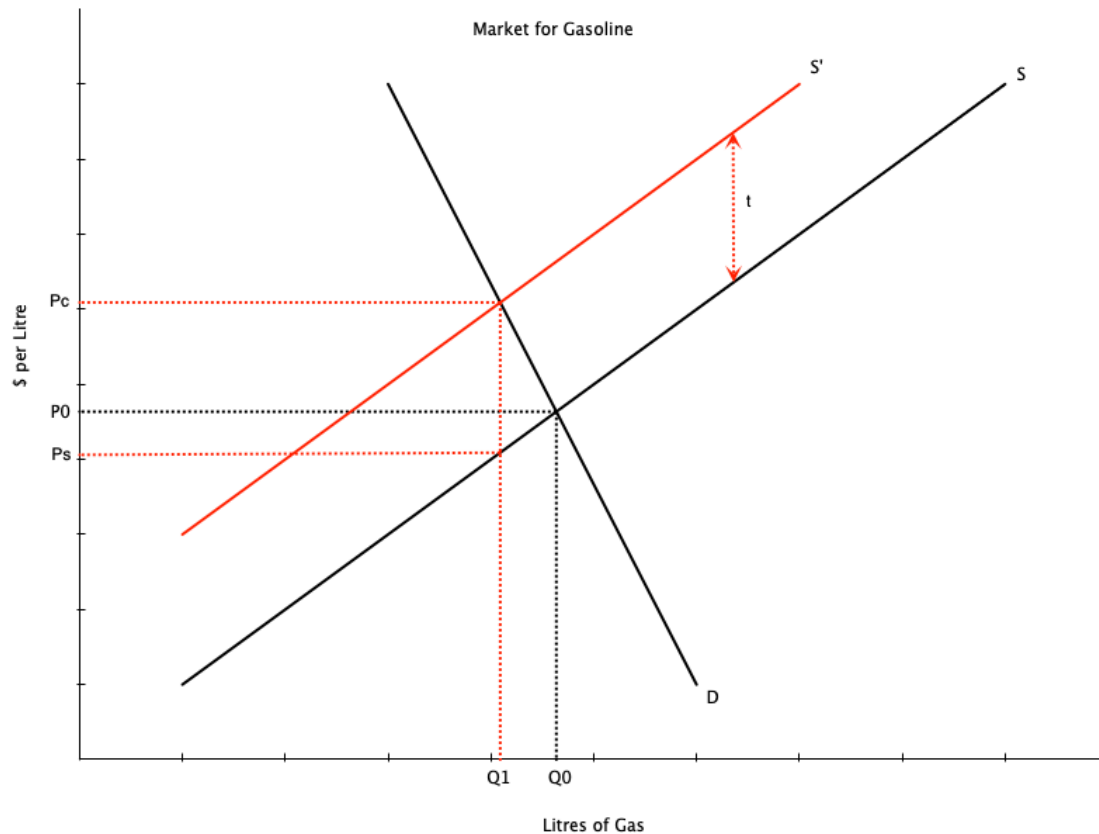


- The tax introduces a **tax wedge**
  - Difference between what consumers pay and what producers receive from a transaction
- Consumers pay the equilibrium price  $P_c$ 
  - Price includes the tax
- Producers receive  $P_s = P_c - t$ 
  - They remit  $t$  for every unit sold
- Quantity falls to  $Q_1$

# Unit Tax on Sellers - Graphical

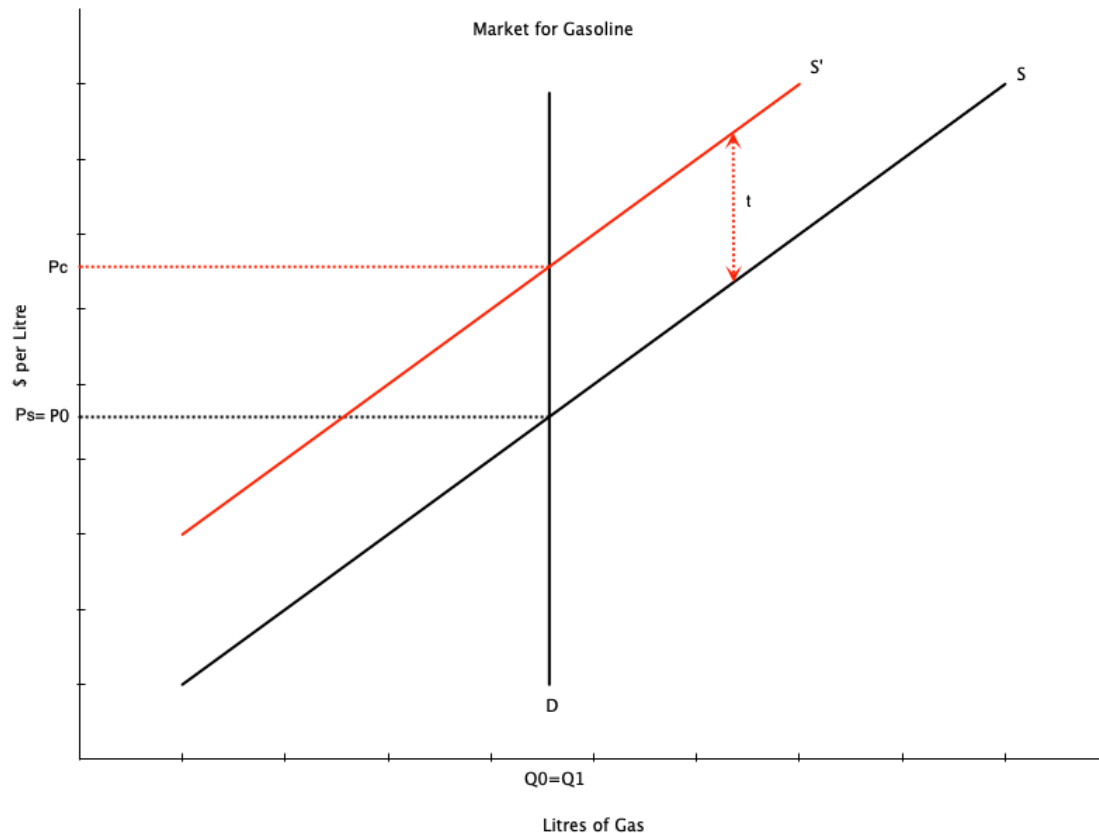
- In example above, economic incidence is shared equally between consumers and producers
  - Consumers pay  $P_c - P_0$  more than before tax
  - Producers receive  $P_0 - P_s$  less than before tax
  - These amounts are equal
- Equal economic incidence is specific to this example because supply and demand have the same slope
- In general, **economic incidence depends on the elasticities of supply and demand**
  - A **more elastic demand** curve means consumers bear less of the economic incidence
    - Higher elasticity means consumers can switch to other goods when price changes
  - A **more elastic supply** curve means producers bear less of the economic incidence
    - A firm with higher elasticity can alter production easily when prices change

# Unit Tax on Sellers - Graphical



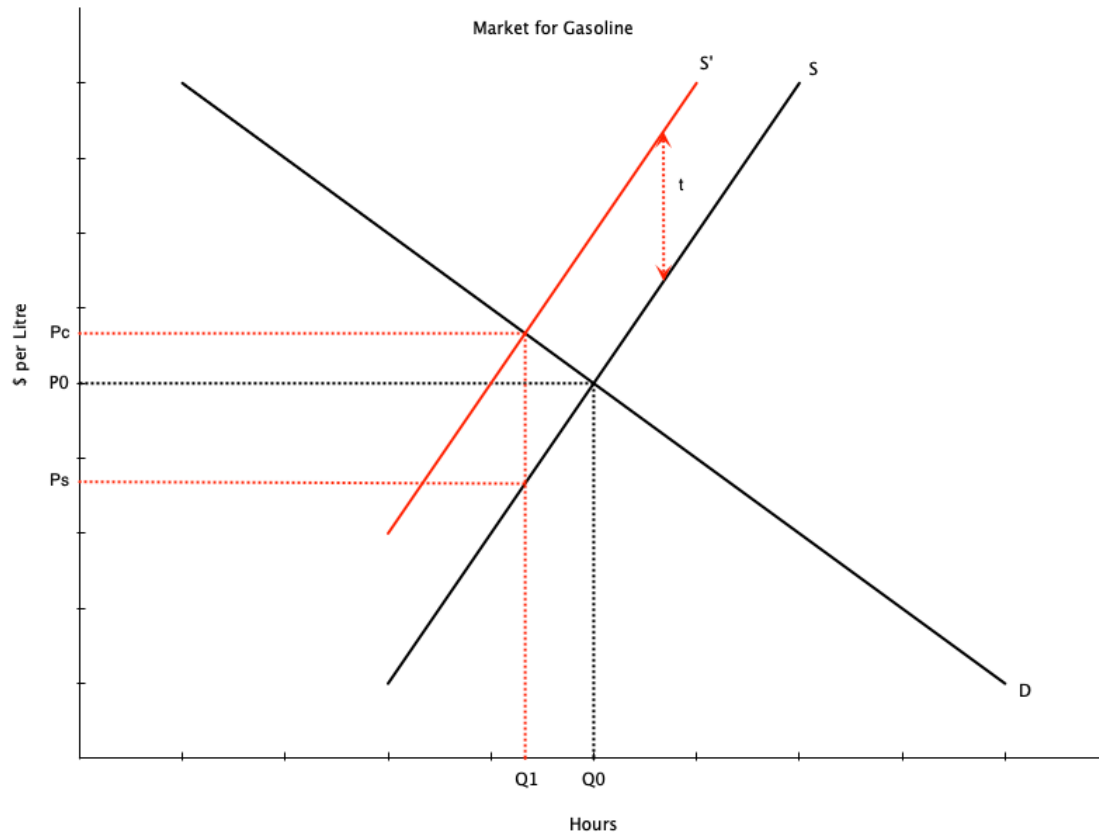
- Graph to the right shows a more **inelastic demand** curve
- After tax consumers pay  $P_c$
- Producers receive  $P_s = P_c - t$
- But  $P_c - P_0$  is now larger than  $P_0 - P_s$ 
  - Inelastic demand means consumers less able to substitute
  - They absorb more of the economic incidence

# Unit Tax on Sellers - Graphical



- In extreme with **perfectly inelastic demand**, consumers bear entire economic incidence
  - Perfectly inelastic demand means complete inability to substitute
- After tax consumers pay  $P_c = P_0 + t$
- Producers receive  $P_s = P_c - t = P_0$
- No change in quantity
  - Since consumers demand  $Q_0 = Q_1$  at any price

# Unit Tax on Sellers - Graphical



- Now imagine a more **inelastic supply** curve
  - Firms less able to adjust quantities when prices change
- After tax consumers pay  $P_c'$
- Producers receive  $P_s' = P_c' - t$
- $P_0 - P_s'$  is larger than  $P_c' - P_0$ 
  - Firms absorb more of the economic incidence
  - They are less able to adjust production to avoid the tax

# Unit Tax on Sellers - Math

- We can show the same result mathematically
- Use linear inverse demand and supply curves with clean numbers

$$\text{Demand: } P_c = 14 - Q_c, \quad \text{Supply: } P_s = 2 + Q_s$$

- Equilibrium without tax is where  $P_s = P_c = P_0$ ,  $Q_c = Q_s = Q_0$

$$14 - Q_0 = 2 + Q_0$$

$$12 = 2Q_0 \Rightarrow Q_0 = 6$$

- Sub  $Q_0$  into either equation to get  $P_0$

$$P_0 = 14 - 6 = 8$$

# Unit Tax on Sellers - Math

- Now introduce a per unit tax  $t = 4$  on sellers
- In equilibrium,  $P_s + t = P_c$  and  $Q_c = Q_s = Q_1$ 
  - There is a wedge between what consumers pay and what producers receive
- Substituting in the equations for demand and supply

$$14 - Q_1 = 2 + Q_1 + 4$$

$$14 - Q_1 = 6 + Q_1 \Rightarrow Q_1 = 4$$

- Sub  $Q_1$  into either equation to get  $P_c$  or  $P_s$

$$P_c = 14 - 4 = 10$$

$$P_s = 2 + 4 = 6$$

# Unit Tax on Sellers - Math

- Key things to take away when the tax is introduced
  - Quantity falls from  $Q_0 = 6$  to  $Q_1 = 4$
  - Consumers pay  $P_c = 10$ , which is  $10 - 8 = 2$  more than before tax
  - Producers receive  $P_s = 6$ , which is  $8 - 6 = 2$  less than before tax
  - Economic incidence is shared equally between consumers and producers
- Example is specific to when demand and supply have the same slope
- In general, economic incidence depends on elasticities of supply and demand

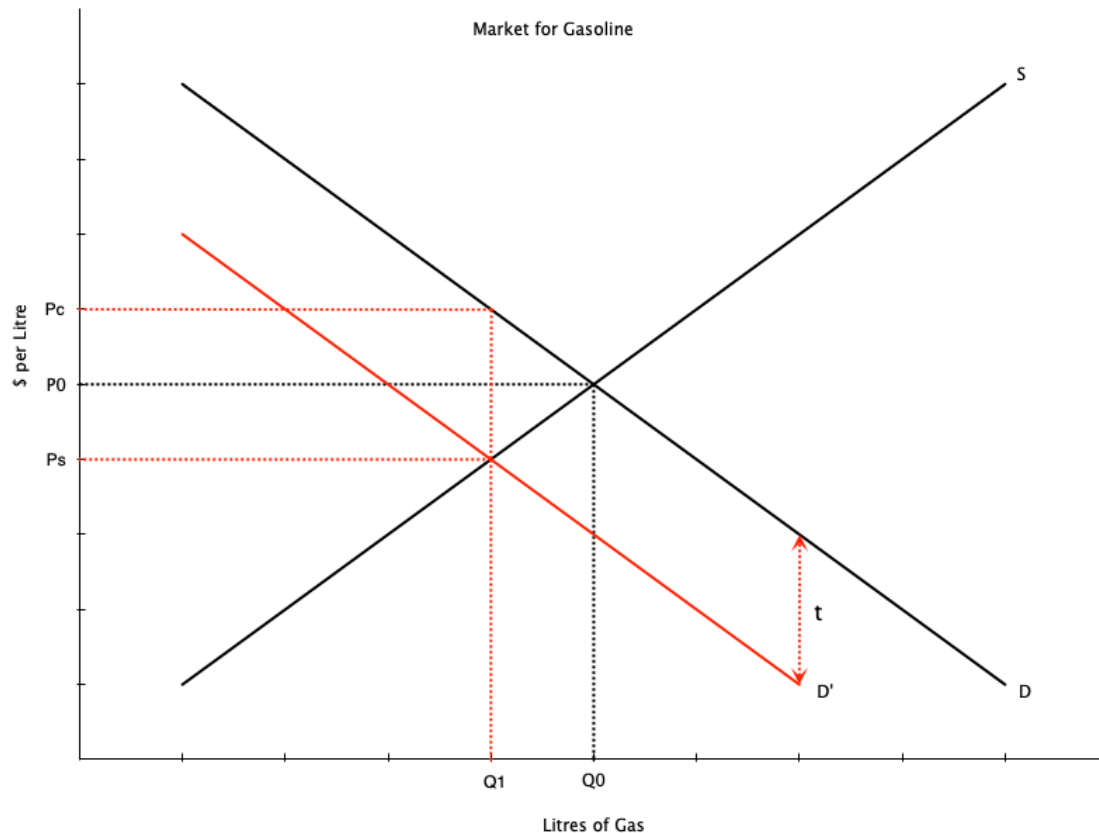
$$\Delta P_c = P_c - P_0 = \frac{\varepsilon_s}{\varepsilon_s + |\varepsilon_d|} t, \quad \Delta P_s = P_0 - P_s = \frac{|\varepsilon_d|}{\varepsilon_s + |\varepsilon_d|} t$$

- $\varepsilon_s = \frac{dQ_s}{dP_s} \frac{P}{Q}$  is elasticity of supply and  $\varepsilon_d = \frac{dQ_c}{dP_c} \frac{P}{Q}$  is elasticity of demand

# Unit Tax on Buyers - Graphical

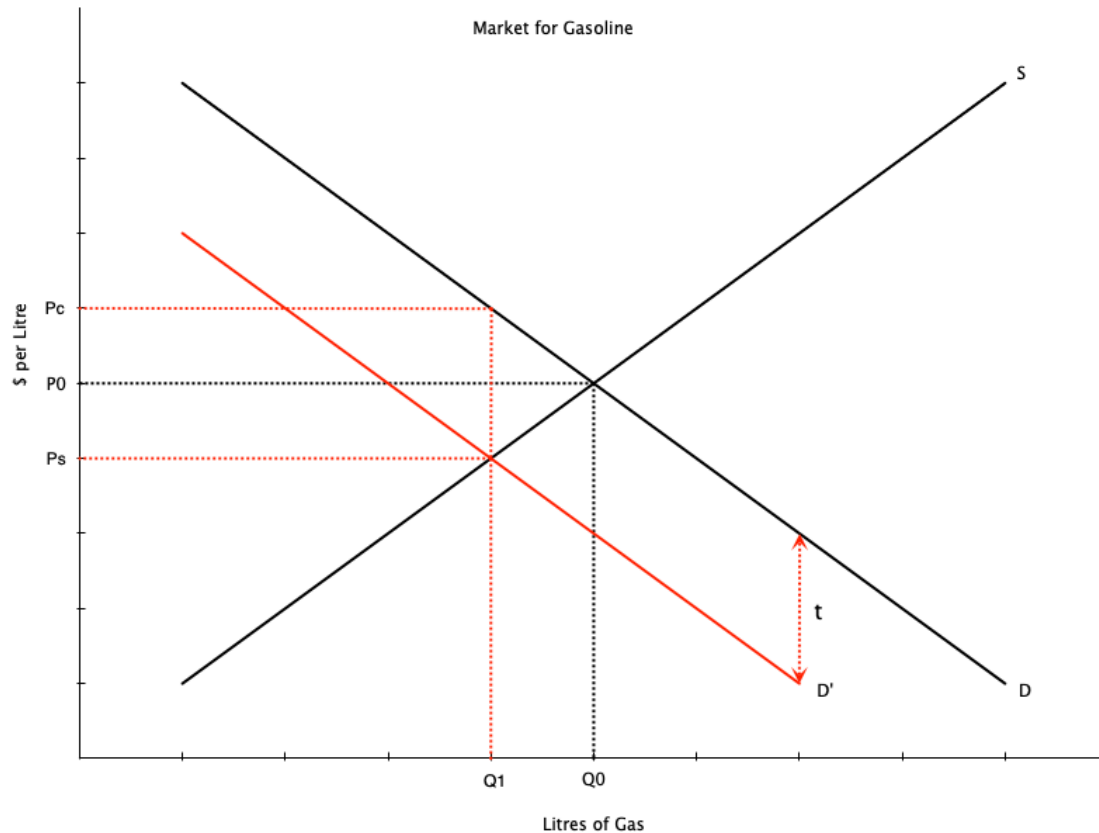
- More rarely, taxes are levied on buyers of goods and services
- We can use the same graphical tools to examine this case
- Key lesson is that **economic incidence does not depend on whether the tax is levied on buyers or sellers**

# Unit Tax on Buyers - Graphical



- Now government levies a per unit tax  $t$  (e.g. \$0.10/litre) on gasoline
- Assume gas tax is levied on **buyers** (statutory incidence)
- This shifts the demand curve down by the amount of the tax
  - New demand curve is  $D'$
  - At each quantity, buyers want to pay  $t$  less to cover the tax
- New equilibrium is where  $S$  intersects  $D'$ 
  - Determines price received by the seller

# Unit Tax on Buyers - Graphical



- Notice that  $P_s$  and  $P_c$  are the same as when the tax was levied on sellers
  - Producers receive  $P_s$
  - Consumers pay  $P_c = P_s + t$
  - Quantity falls to  $Q_1$
- In this case, the slopes are equal so the burden is shared
- Changes in elasticities affect economic burden in the same way

# Unit Tax on Buyers - Math

- The math is exactly the same as when the tax is levied on sellers

$$\text{Demand: } P_c = 14 - Q_c, \quad \text{Supply: } P_s = 2 + Q_s$$

- Tax is  $t = 4$  on buyers
- In equilibrium,  $P_s = P_c - t$  and  $Q_c = Q_s = Q_1$
- Algebra is same as we did above, so we get the same results

$$P_c = 10$$

$$P_s = 6$$

# Unit Taxes - Takeaways

## Three Key Lessons

1. Statutory incidence does not determine economic incidence
  2. Economic incidence depends on elasticities of supply and demand
  3. Economic incidence does not depend on whether the tax is levied on buyers or sellers
- More elastic demand means consumers bear less of the economic incidence
  - More elastic supply means producers bear less of the economic incidence

# iClicker Question

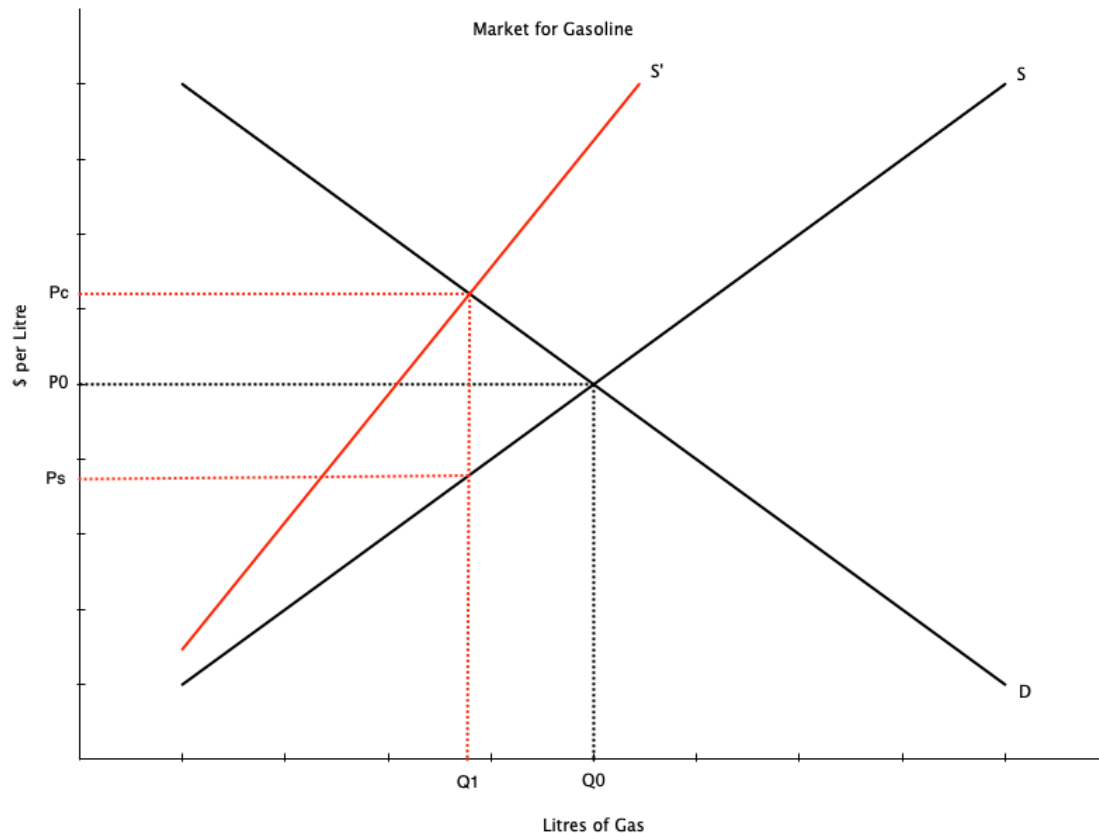
Gasoline demand is quite inelastic (about  $-0.2$ ) while supply is elastic (about  $2.0$ ). If a \$1-per-litre tax is levied on sellers, who bears most of the burden?

- A) Consumers — the less elastic side of the market bears more
- B) Sellers — the tax is levied on them
- C) Sellers — the more elastic side of the market bears more
- D) It is split 50/50
- E) No one — the tax only reduces the quantity sold

# Ad Valorem Taxes

- So far we have only considered **unit taxes**
  - A fixed amount per unit sold (e.g. \$0.10/litre)
- More common are **ad valorem taxes**
  - A percentage of the price (e.g. 13% HST on most goods and services)
- Ad valorem taxes create a tax wedge that increases with the price
  - Higher priced goods have a larger tax wedge
- As before
  - Economic incidence depends on elasticities of supply and demand
  - Economic incidence does not depend on whether the tax is levied on buyers or sellers

# Ad Valorem Taxes

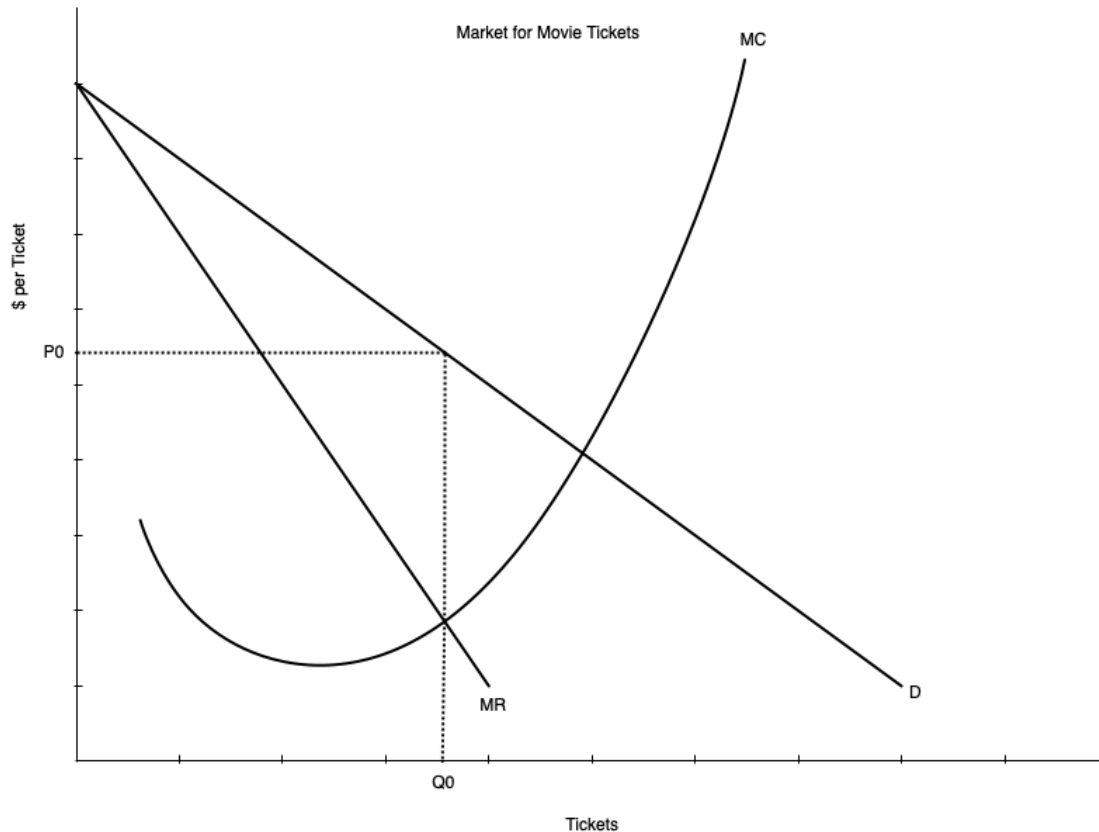


- Analysis is similar to unit tax except supply curve shifts up and becomes steeper
- If tax is levied on buyers, demand curve shifts down and becomes flatter
- Still a wedge between what consumers pay and what producers receive
- Economic burden on sellers decreases with supply elasticity
- Economic burden on buyers decreases with demand elasticity

# Unit Tax on a Monopolist

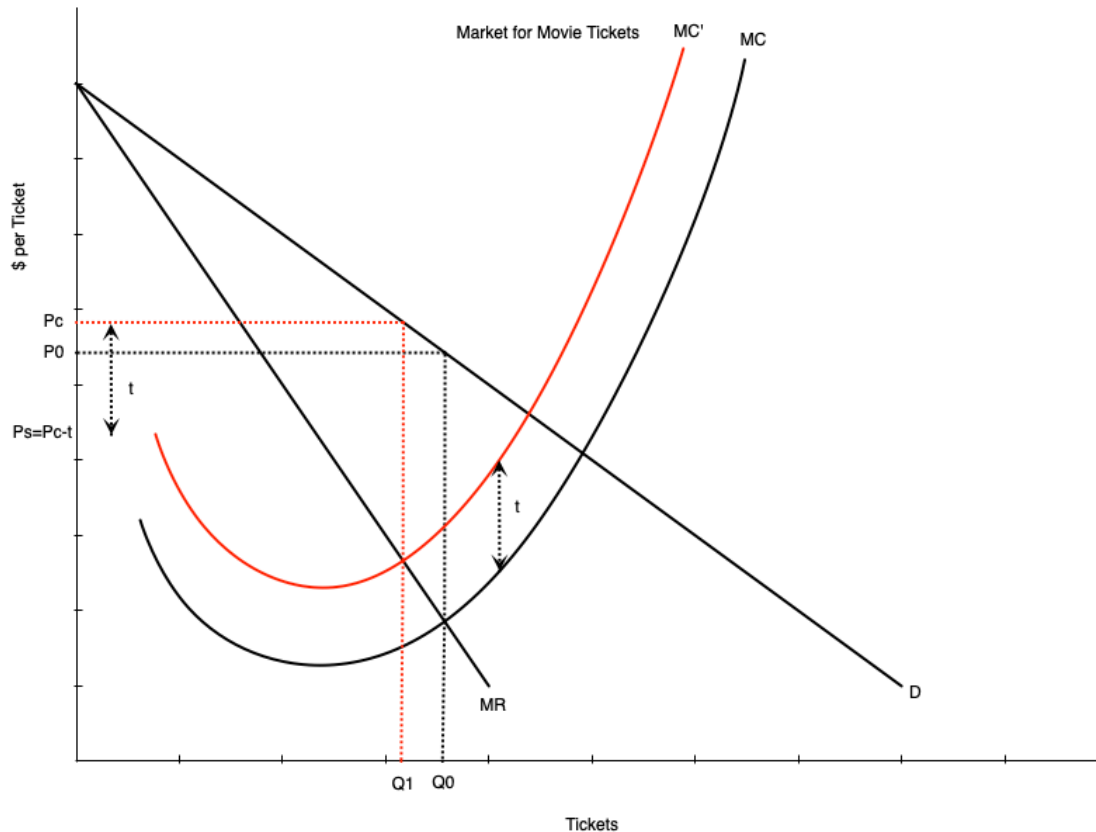
- So far we have only considered perfectly competitive markets
- What happens when there is only one seller in the market?
- A tax increases marginal cost, so the monopolist produces less
- Effects of the tax are potentially different from competitive markets
  - Quantity will fall
  - Price paid by consumers will rise
  - But, price received by the monopolist may rise or fall
    - Depends on cost structure and demand elasticity

# Unit Tax on a Monopolist



- Graph shows pre tax equilibrium in a **monopoly**
- ATC curve omitted for graph clarity
- Example: market for movie tickets
  - Dominated by a couple of firms
  - Not strictly a monopoly but close enough
- Equilibrium is where  $MR = MC$ 
  - Price is  $P_0$
  - Quantity is  $Q_0$

# Unit Tax on a Monopolist



- Government levies a per unit tax  $t$  on each ticket sold
- This shifts the cost curves up by the amount of the tax
  - New marginal cost is  $MC'$
- Equilibrium is where  $MR = MC'$ 
  - Price paid by consumers is  $P_c$
  - Price received by monopolist is  $P_s = P_c - t$
  - Quantity is lower at  $Q_1$
- Consumers bear more of the tax if
  - Demand is **inelastic**
  - Marginal cost is relatively flat

# iClicker Question

A unit tax is levied on a monopolist. Since it is the only seller in the market, can it simply pass the entire tax on to consumers?

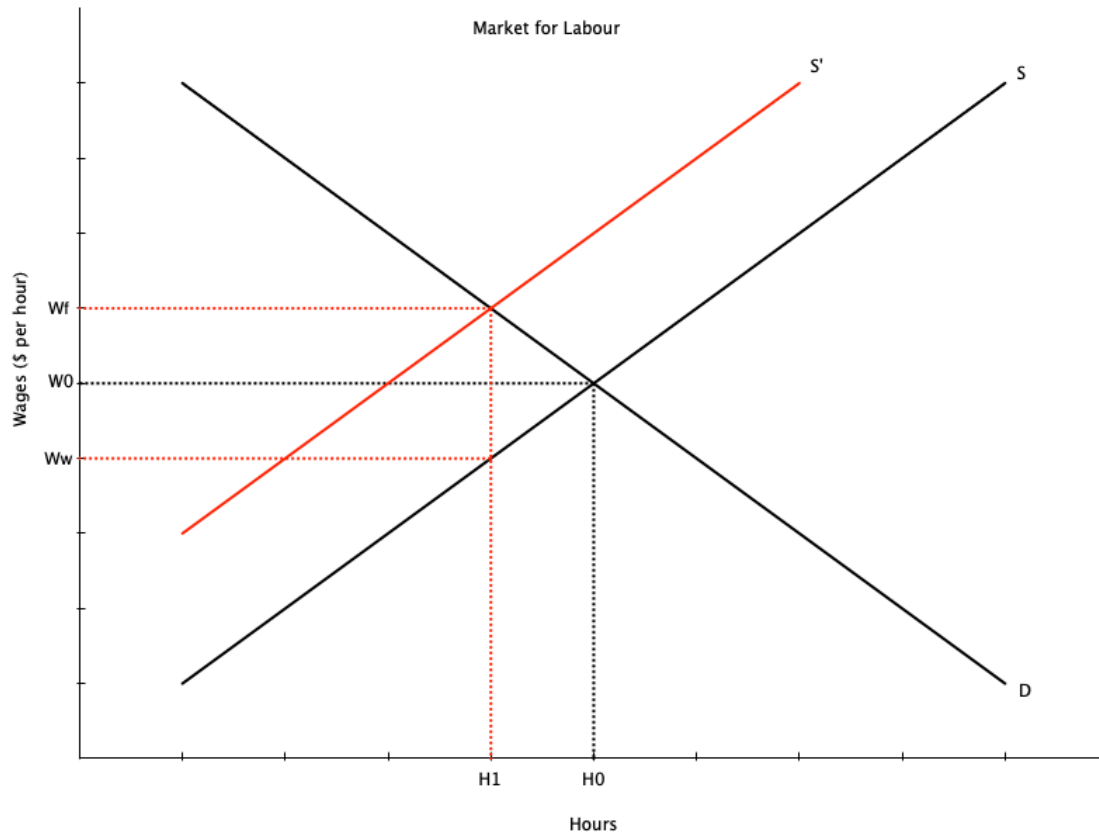
- A) No — price typically rises by less than the tax, so the monopolist bears part of the burden
- B) Yes — market power lets it pass on the full tax
- C) No — the monopolist must absorb the entire tax, because its price is already as high as possible
- D) Yes — but only if demand is elastic
- E) The tax leaves price and profits unchanged

# Taxes on Factor Markets in Partial Equilibrium Models

# Introduction

- So far we have only considered taxes on goods and services
- Taxes can also be levied on **factors of production**
  - Labour
  - Capital
- Taxes affect prices paid for and received by those factors
  - Example: payroll taxes affect wages paid and received
- Taxes also affect quantities of factors employed
- Analysis is the same as taxes on goods and services
  - Only difference is that firms are buyers and households are sellers

# Payroll Tax on Workers



- Suppose government levies a **payroll tax**  $t$  on workers
  - Example: Employment Insurance (EI) premiums paid by employees
- This shifts the supply curve of labour up by the amount of the tax
  - At each quantity, workers want to be paid  $t$  more to cover the tax
- Creates wedge between what workers are paid and what they keep
  - Firms now pay  $W_f$
  - Workers receive  $W_w = W_f - t$

# Payroll Tax on Workers

- Incidence again depends on supply and demand elasticity
- A more **inelastic supply** curve means workers bear more of the economic incidence
  - They are less able to change work hours when wages change
  - Makes them stuck in a job, so they bear more of the tax
- A more **inelastic demand** curve means firms bear more of the economic incidence
  - They are less able to substitute between labour and other inputs
  - Makes number of workers inflexible, so they bear more of the tax
- Does not matter if tax is levied on workers or firms
  - In reality payroll taxes are often levied on both

# iClicker Question

CPP premiums are split 50/50 between employers and employees by law. If labour supply is quite inelastic in the long run, who actually bears the employer half?

- A) Mostly workers, through lower wages
- B) Mostly employers — the statutory split determines the burden
- C) Mostly consumers, through higher prices
- D) The government
- E) Exactly 50/50, matching the statutory split

# Capital Taxation

- Labour is not the only input into production
- Capital is also used
  - For this analysis, think of capital as the money used to finance machines, factories, etc
- Capital owners earn a return on investments, and that can be taxed
- The analysis of taxing capital is the same as for labour
  - Except that the rate of return ( $r$ ) is the price
  - Total investment (dollars) is the quantity
  - Firms are the buyers, people with money are the suppliers

# Capital Taxation

- A key consideration in the analysis is **capital mobility** across countries
  - The ease with which people can move money and invest around the globe
- If capital is **perfectly mobile**, the buyers of capital bear the entire burden
  - Perfect mobility means a perfectly elastic supply curve
  - The price is set at the global rate of return
  - When supply is perfectly elastic, buyers bear the whole tax burden
- In reality, capital is mobile but not perfectly
  - Buyers and sellers are likely to bear some of the burden of capital taxation

# Taxes on Profits and Land

- Governments can also levy taxes on economic profits
  - **Economic profits**: profits above a normal rate of return
- These types of taxes are borne entirely by firms
  - Maximizing untaxed profits and taxed profits yields the same prices and quantities
- Consider a monopoly (because this would be a market where economic profits happen)
- Without taxes, maximize

$$\Pi(q) = p(q)q - C(q) = (a - bq)q - cq = (a - c)q - bq^2$$

- Take the derivative with respect to  $q$

$$(a - c) - 2bq = 0$$

# Taxes on Profits and Land

- Solve for  $q$  to get the profit maximizing quantity

$$q^* = \frac{a - c}{2b}$$

- Price is then

$$p^* = a - bq^* = \frac{a + c}{2}$$

- With taxes, maximize

$$(1 - t)\Pi(q) = (1 - t)(p(q)q - C(q)) = (1 - t)((a - c)q - bq^2)$$

# Taxes on Profits and Land

- Take the derivative with respect to  $q$

$$(1 - t)((a - c) - 2bq) = 0$$

- Which leads to the same prices and quantities
- Means that the economic incidence of profit taxes is entirely on firms
  - Profits go from  $\Pi(q^*)$  to  $(1 - t)\Pi(q^*)$
- Problem with these taxes is that they are **hard to implement** in practice

# Taxes on Land

- Historically there were taxes on only land
  - As opposed to modern property taxes that also tax buildings
- When land is taxed the incidence is entirely on current owners
  - The value of the land drops by the amount of the current and future tax obligations
- Can see this by looking at the value for a piece of land
- Without taxes, the value of land is the present value of future rents

$$P_R = \sum_{t=0}^T \frac{R_t}{(1+r)^t}$$

# Taxes on Land

- If there is a tax  $u_t$  on land, the value becomes

$$P'_R = \sum_{t=0}^T \frac{R_t - u_t}{(1+r)^t} = \sum_{t=0}^T \frac{R_t}{(1+r)^t} - \sum_{t=0}^T \frac{u_t}{(1+r)^t}$$

- The difference between  $P_R$  and  $P'_R$  is the present value of all future tax obligations

$$P'_R - P_R = - \sum_{t=0}^T \frac{u_t}{(1+r)^t}$$

- When the tax is levied, the value of the land immediately drops by the tax obligation
  - So it is borne by the current owner

# iClicker Question

A city unexpectedly announces a permanent increase in the tax on land. Who bears the burden?

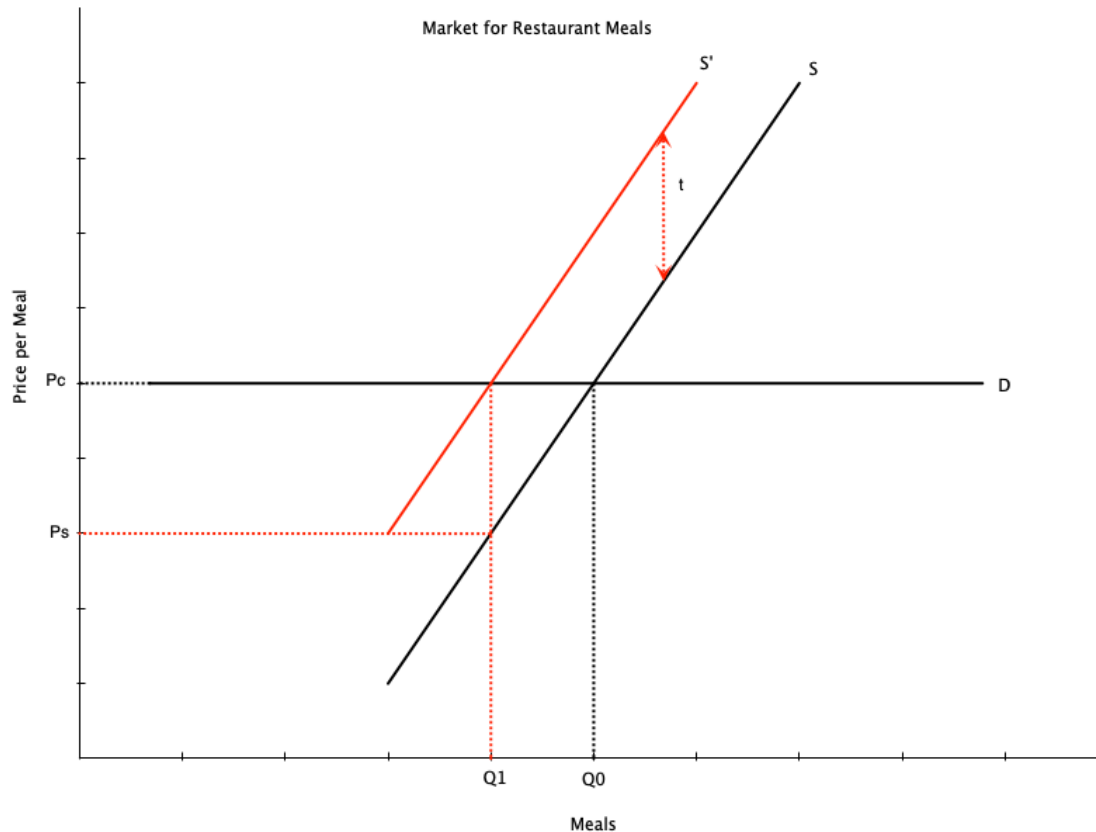
- A) The owners at the time of the announcement — the land price falls by the present value of the future taxes
- B) All current and future owners equally, since the tax is paid every year
- C) Tenants, through higher rents
- D) Nobody — landowners simply raise rents to cover it
- E) Future buyers, since they will be the ones paying the annual tax bills

# General Equilibrium

# Introduction

- So far we have only considered **partial equilibrium** models
  - One market at a time
- It is informative, but incomplete
- In reality, markets are interconnected
  - A tax in one market can affect other markets
  - Example: a tax on goods can affect labour market
    - If people buy less, firms produce less
    - Firms hire fewer workers
    - Labour market is affected
- **General equilibrium** models consider all markets simultaneously

# A Tax on Goods and Services

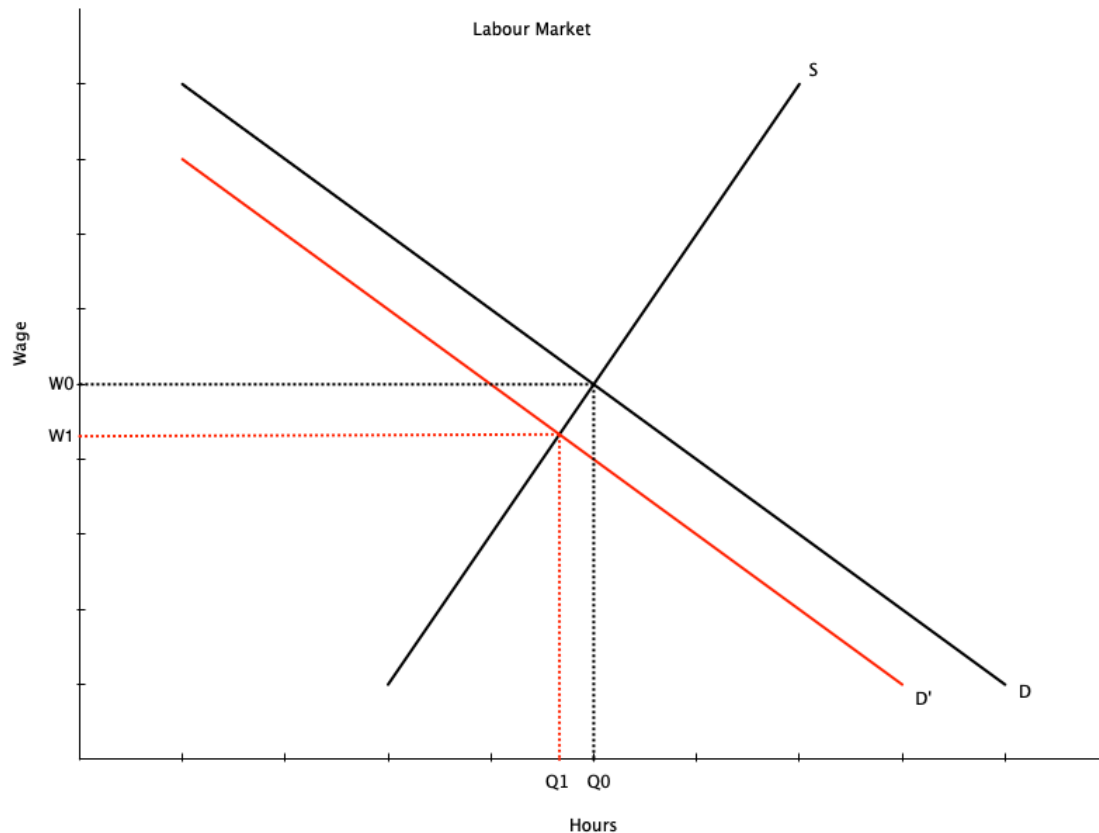


- Consider a unit tax on restaurants
- For simplicity, assume restaurant meals have perfectly elastic demand
  - People are very sensitive to the price and will cook at home if price rises
- The full burden of the tax is on the restaurants
- Quantity of meals falls because of the tax

# A Tax on Goods and Services

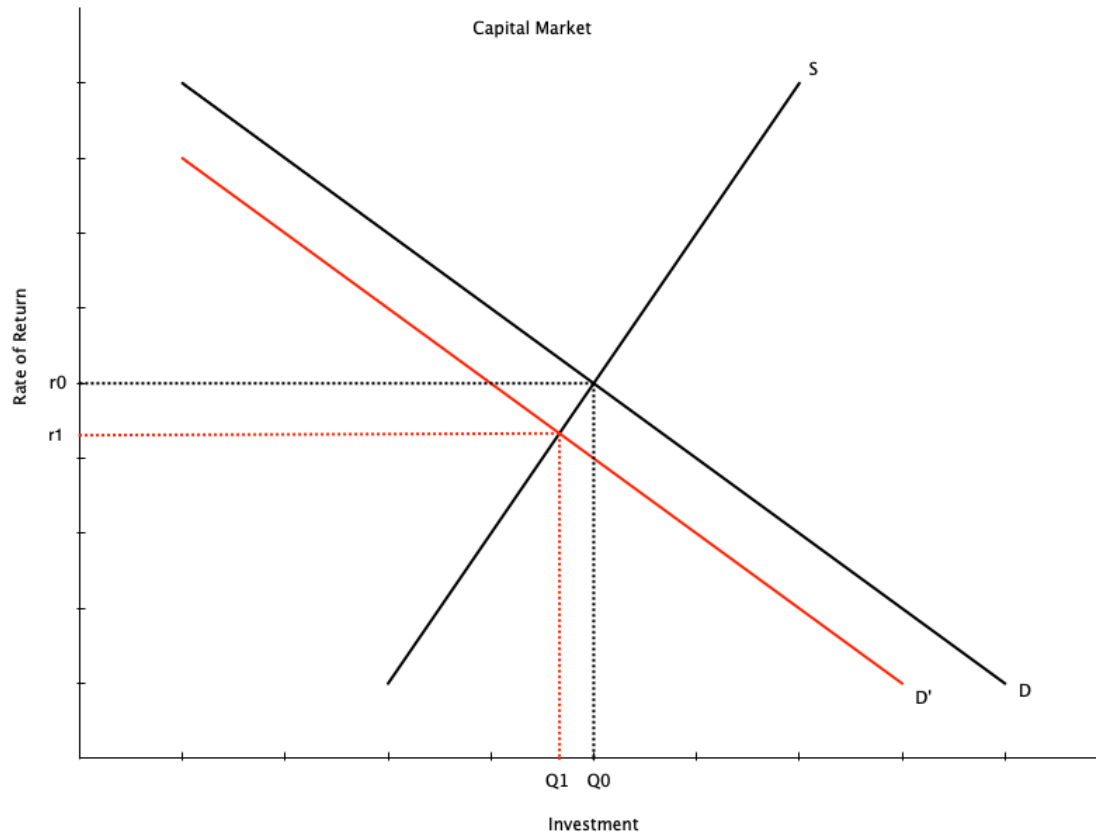
- But a restaurant cannot bear taxes
- A restaurant is really a production process that combines labour and capital
  - A production process cannot pay a tax
- It is ultimately the **workers and owners of capital** that pay the tax
- The tax on meals will filter down to the labour and capital markets
  - Workers will get paid less
  - Owners of capital will earn less

# A Tax on Goods and Services



- Consider the labour market for restaurant workers
- The tax on meals shifts the demand for restaurant workers to the left
  - Labour demand is derived partly from the demand for meals
  - Less demand for meals means less demand for workers
- The equilibrium wage falls and workers are paid less
- Part of the burden falls on workers in the form of **lower wages**

# A Tax on Goods and Services



- Similar things happen in the capital market
- The tax on meals shifts the demand for capital to the left
  - Less demand for meals means less demand for capital
- The equilibrium return to capital falls and owners are paid less
- Part of the burden falls on owners of capital in the form of **lower returns**

# A Tax on Goods and Services

- The degree to which workers and owners of capital bear the burden depends on elasticities
  - More **elastic labour supply** means workers bear less of the burden
  - More **elastic capital supply** means owners of capital bear less of the burden
- In our example, we assumed perfectly elastic demand for meals
  - This meant that restaurants bore the entire burden of the tax
- If consumer demand were less elastic
  - Consumers would bear some of the burden
  - Would affect other goods they consume
    - Increase consumption of substitutes
    - Decrease consumption of complements
    - Decrease consumption of all goods via an income effect

# A Tax on Goods and Services

- General equilibrium analysis can get very complicated
  - Many interconnected markets
  - Changes in one market affect many others
- Can affect markets vertically and horizontally
  - Vertically: factors of production
  - Horizontally: related goods (e.g. tax on coffee affects tea market)
- Main lesson is still the same: **statutory incidence does not determine economic incidence**

# Tax Equivalencies

- Two taxes are **equivalent** if they have the same economic incidence
  - They change the same relative prices, so the same people end up bearing the burden
- Useful for policy: taxes that look very different can place the same burden on the same people
- To see the logic, consider a simple economy with
  - Two goods: food ( $F$ ) and manufacturing ( $M$ )
  - Two factors of production: labour ( $L$ ) and capital ( $K$ )

# Tax Equivalencies

- In this economy, a government could levy nine different taxes

|                      | Tax on capital | Tax on labour | Tax on the good  |
|----------------------|----------------|---------------|------------------|
| Food sector          | $t_{KF}$       | $t_{LF}$      | $t_F$            |
| Manufacturing sector | $t_{KM}$       | $t_{LM}$      | $t_M$            |
| Both sectors         | $t_K$          | $t_L$         | $t$ (income tax) |

- Taxes on one factor in one sector ( $t_{KF}$  ,  $t_{LF}$  ,  $t_{KM}$  ,  $t_{LM}$  ) are called **partial factor taxes**

# Tax Equivalencies

- Across a row: taxing both factors in a sector  $\equiv$  taxing that sector's output:  $t_{KF} + t_{LF} \equiv t_F$
- Why? Under perfect competition, every dollar of food revenue is paid out to the factors that produce it
  - For every \$100 of food sold, the firm pays out \$60 to labour and \$40 to capital
  - A 10% tax on both factor payments collects \$10 per \$100 of food — exactly what a 10% tax on food collects
  - Buyers, workers, and capital owners face the same tax wedge either way, so the incidence is identical

# Tax Equivalencies

- Down a column: taxing one factor in both sectors  $\equiv$  a general tax on that factor:  
$$t_{KF} + t_{KM} \equiv t_K$$
- Why? If capital faces the same tax rate everywhere, switching sectors cannot avoid the tax
  - Nothing reallocates between sectors — the net return to capital simply falls
  - Contrast with a partial factor tax like  $t_{KF}$  alone, where capital *can* escape into manufacturing

# Tax Equivalencies

- All together: taxing all goods  $\equiv$  taxing all factors  $\equiv$  an income tax:  $t_F + t_M \equiv t_K + t_L \equiv t$
- Why? All income is earned from capital or labour, and in this model all income is spent on the two goods
  - A tax on everything you earn and a tax on everything you buy hit the same base
- So we only need to work out the incidence of a few basic taxes to understand all nine
  - A tax on one good ( $t_F$ ) and a partial factor tax ( $t_{KF}$ ) — exactly the cases the Harberger model studies

# iClicker Question

In the two-good, two-factor model, a tax on capital in the food sector combined with a tax on labour in the food sector ( $t_{KF} + t_{LF}$ ) is equivalent to which single tax?

- A) A tax on food output ( $t_F$ )
- B) A general tax on capital ( $t_K$ )
- C) A general tax on labour ( $t_L$ )
- D) An income tax ( $t$ )
- E) There is no equivalent single tax

# Harberger Model

- [Arnold Harberger](#) studied tax incidence in general equilibrium
- Basics of his model
  - Technology
    - Capital and labour used in production
    - Constant returns to scale
    - Ease of substitution between capital and labour different between sectors
  - Factor suppliers
    - Capital and labour perfectly mobile
    - Returns to both are the same in both sectors
  - Market structure is perfectly competitive
  - Factors are in fixed supply
  - Consumers are all the same

# Harberger Model

- Suppose a tax on food ( $t_F$ ) is introduced, and food is **capital intensive**
  - It uses more capital per worker than manufacturing — say 3 machines per worker vs 1
- Step by step
  - Price of food rises, so consumers buy less food and more manufacturing
    - Harberger assumes the tax revenue is returned to consumers, so this is pure **substitution** between goods — consumers are not poorer overall
  - The shrinking food sector releases capital and labour, which must find work in manufacturing
  - Food releases 3 machines per worker, but manufacturing only wants 1 machine per worker at current prices
  - There is more capital than the market wants: the return to capital must fall until manufacturing willingly absorbs it all

# Harberger Model

## Key Result

A tax on a good falls more heavily on the factor used **intensively** in producing that good

- Here: the food tax lowers the return to capital
- Capital is mobile, so returns equalize across sectors
  - Capital owners in *both* sectors bear the burden, not just those in food
- Strength of these effects depends on elasticities
  - How easily consumers switch to manufacturing, and how easily sectors substitute between labour and capital

# Harberger Model

- Suppose an income tax ( $t$ ) is introduced
  - Use equivalencies table
  - Income tax is equivalent to a combination of taxes on capital and labour
  - Both are in fixed supply (**perfectly inelastic**) so they bear the whole burden
- A tax on labour ( $t_L$ ) is introduced
  - Tax on labour in both sectors
  - Labour supply is **perfectly inelastic** so it bears the whole burden
  - No incentive to switch between sectors

# Harberger Model

- Finally, a partial factor tax on capital in manufacturing ( $t_{KM}$ )
  - Price of manufacturing rises
    - Use less capital and labour
    - As capital and labour leave, changes in their relative prices depend on how much is used
  - Producers of manufactured goods use less capital and more labour
    - Demand for capital falls, and so does relative price
    - Price of labour rises
  - Cannot tell what the ultimate effect is on capital or labour prices

# iClicker Question

In the Harberger model, food production is capital-intensive. If a tax is placed on food ( $t_F$ ), who tends to bear the burden in general equilibrium?

- A) Owners of capital in both sectors — the return to capital falls economy-wide
- B) Only owners of capital in the food sector
- C) Only consumers of food
- D) Workers in the manufacturing sector
- E) No one — capital moves to manufacturing and escapes the tax

# Summary

- Tax incidence is not necessarily contained to the market where the tax is levied
- In general equilibrium, taxes can have far reaching effects
  - Both vertically and horizontally
- A full analysis requires consideration of all markets simultaneously
- The key points to remember are
  - Statutory incidence does not determine economic incidence
  - Economic incidence depends on ability to substitute and avoid the tax

# Quantifying Tax Incidence

# Introduction

- We have gone through the theory of tax incidence
- Using data we can see who bears the burden of various taxes
- Could do this with tax data
  - Statistics Canada maintains the Longitudinal Administrative Databank (LAD)
  - Not easy to use because it is confidential microdata
- Can also use data from the [SPSDM simulator](#)
  - Program maintained by Statistics Canada
  - Snapshot of the Canadian tax and transfer system at a point in time

# Tax Progressiveness

## ! Key Definition

- **Average tax rate**: total taxes paid divided by income
  - **Marginal tax rate**: the tax rate paid on the last dollar of income
- 
- Tax systems are classified by how the average tax rate changes with income
    - **Proportional**: average tax rate is constant at all income levels
    - **Progressive**: average tax rate rises with income
    - **Regressive**: average tax rate falls with income
  - Careful: sometimes these terms are defined using the marginal tax rate instead
    - Whether a tax looks progressive can depend on which definition is used

# iClicker Question

Suppose the income tax equals 20% of all income above \$20,000. A person earns \$50,000, so their tax bill is \$6,000. What is their average tax rate?

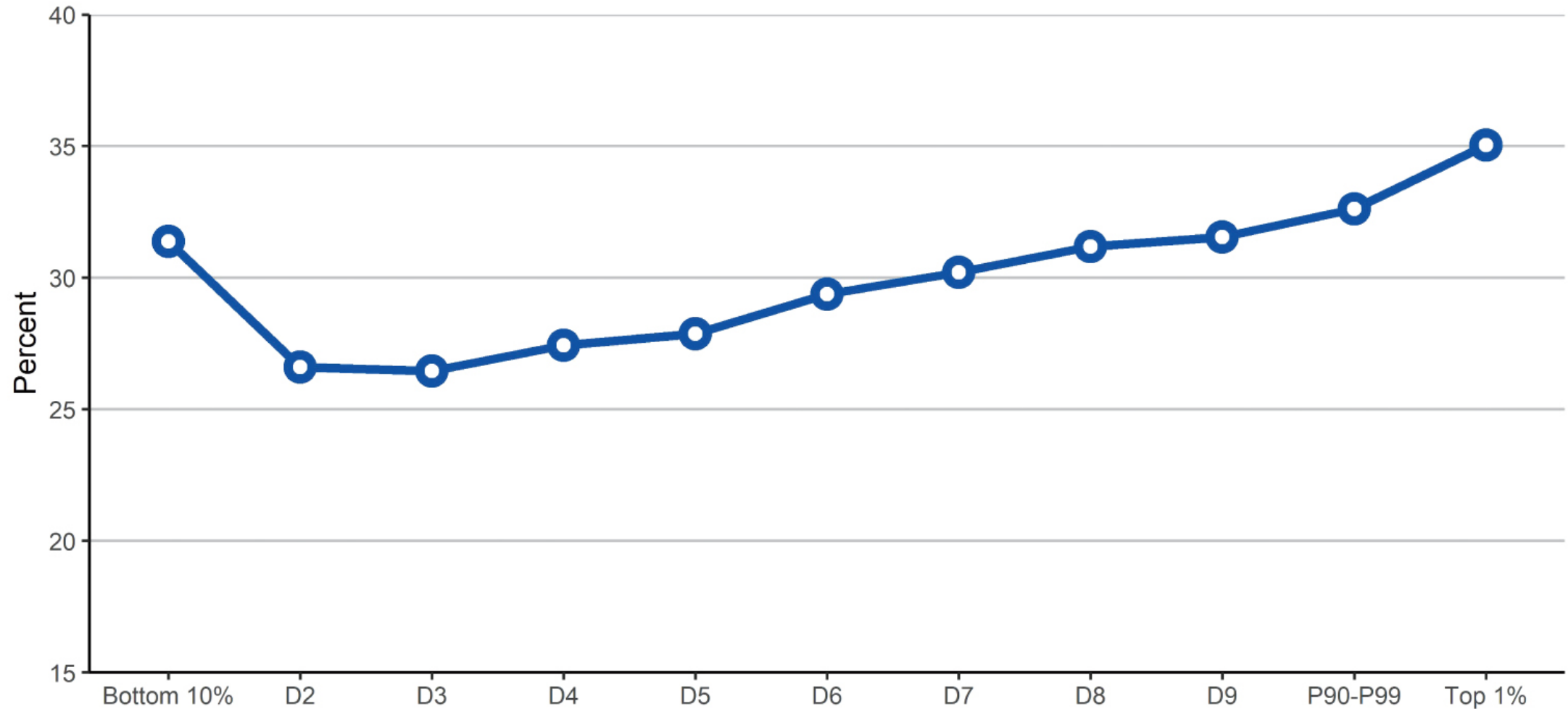
- A) 12%
- B) 20%
- C) 30%
- D) 6%
- E) It cannot be determined without knowing the marginal tax rate

# iClicker Question

Same tax: 20% of all income above \$20,000. Is this tax proportional, progressive, or regressive?

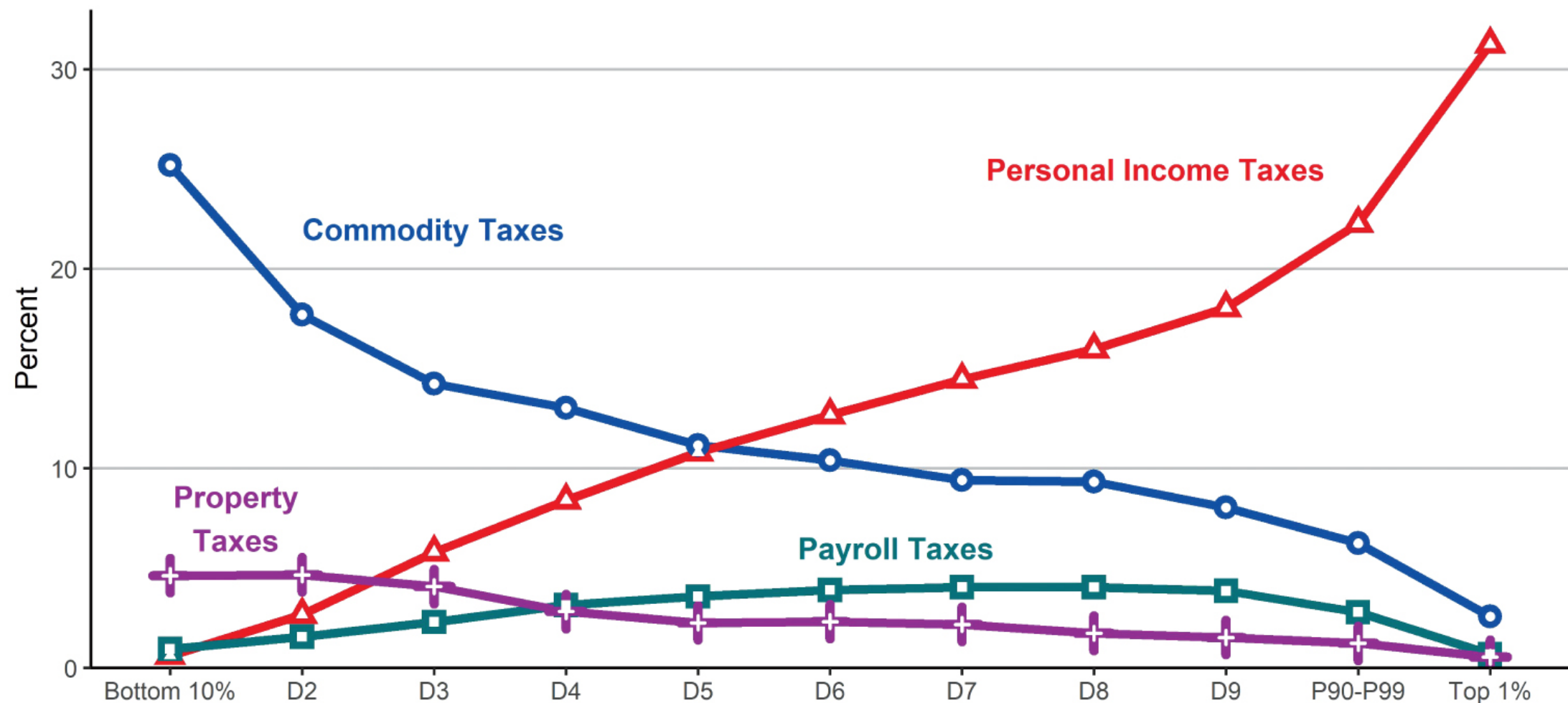
- A) Progressive — the average tax rate rises with income
- B) Proportional — everyone faces the same 20% rate
- C) Regressive — lower earners can least afford to pay
- D) Proportional — the tax is a constant fraction of the tax base
- E) It depends on whether we use the average or the marginal tax rate

# Partial Equilibrium



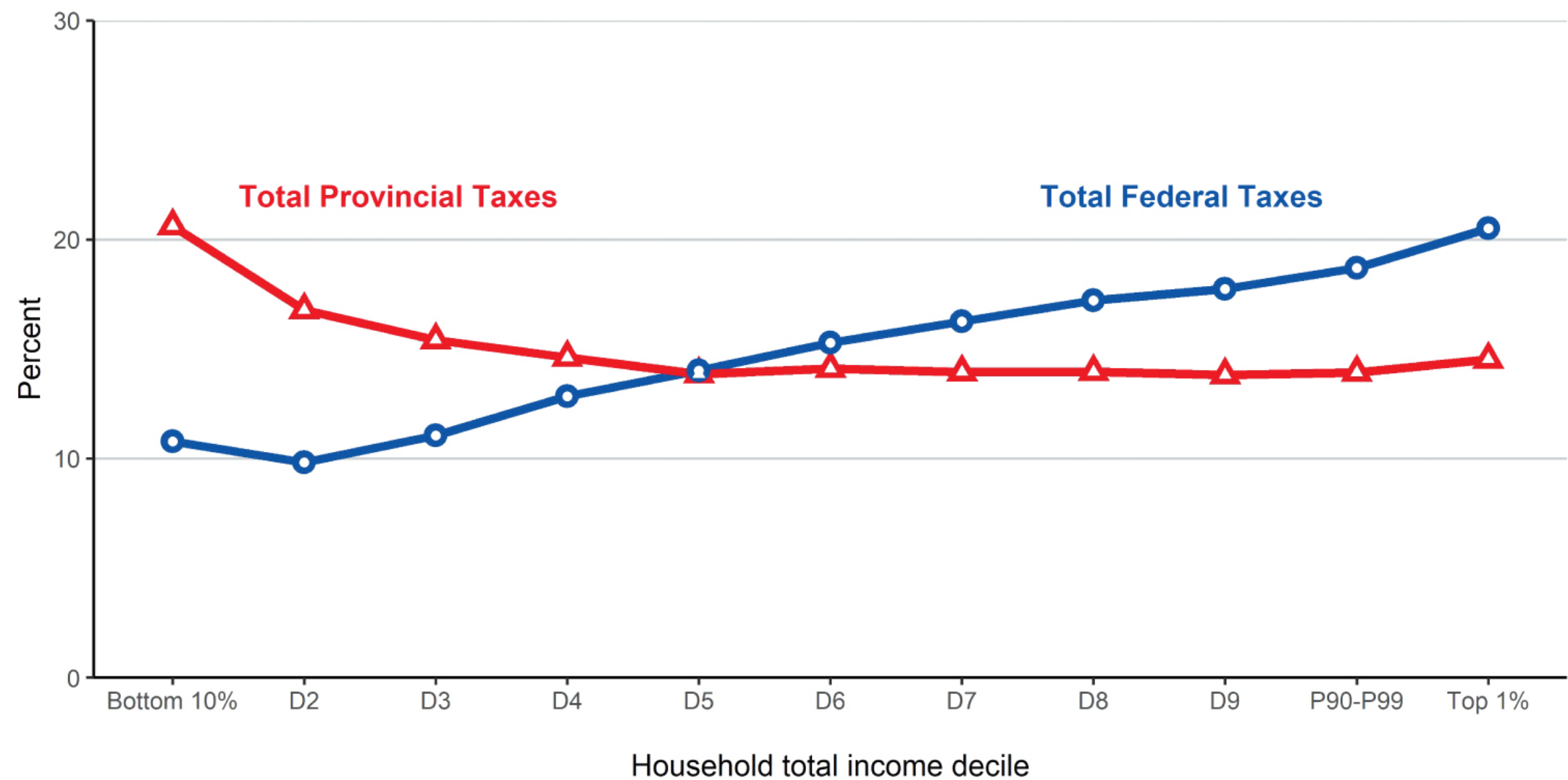
Household total income decile  
Average Tax Rates in Canada

# Partial Equilibrium



Household total income decile  
Average Tax Rates by Source

# Partial Equilibrium



Average Tax Rates by Level of Government

# Partial Equilibrium

- Canada maintains a **progressive** personal income tax system
  - So tax rates rise with income
- Commodity and property taxes are *on their own* **regressive**
  - Tax credits help offset the regressivity of these taxes (e.g. HST credit)
- Payroll taxes are initially **progressive** but turn **regressive**
- Federal taxes are more progressive than provincial taxes
  - Federal taxes mainly come from personal income taxes
  - Provincial taxes contain more commodity and property taxes

# Partial Equilibrium

- Progressivity in these graphs is based on *annual* income
  - Lifetime income might be more appropriate if people consume based on lifetime resources
- Redoing the graphs with lifetime income reduces regressivity

# General Equilibrium

- Evidence on the general equilibrium incidence of taxes is more limited
  - Analysis is more difficult
- Canadian researchers have looked at the tax incidence of the **carbon tax** in BC
  - Use a Computable General Equilibrium (CGE) model
  - A fancy model with many equations you need a computer to solve
- Find that a carbon tax
  - Lowers wages relative to capital income because labour supply is **inelastic**
  - Labour bears more of the burden than capital
  - Tax is **regressive** because wages are a higher share of income for lower income households

# Summary

# Summary

## Key Takeaways

- **Statutory incidence does not determine economic incidence**
- Who “pays” depends on the ability to avoid the tax (elasticities)
- Tax incidence can spread widely across markets in general equilibrium

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